

Software Requirements Specification
for
Gamification event management system
Version 1.0 approved
Prepared by <Omar Khaled>
<AAST>
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Revision History

Name	Date	Reason For Changes	Version

1. Introduction

1.1 Purpose

The gamified event management system is designed to help event organizers create and manage events, while providing a fun and engaging experience for participants. The system will be web-based, making it accessible from anywhere, and will include features such as event creation and management, participant registration and management, task creation and management, task completion tracking, and reward management.

1.2 Document Conventions

Every user should have his unique ID to make it possible to assign points for him (applying concept of gamification)

The admin should have admin control panel

1.3 Intended Audience and Reading Suggestions.

development team, stakeholders, and anyone involved in the planning, design, and implementation of the system. It is important that this document be clear, concise, and accurate, in order to ensure that the system meets the needs of its users and is delivered on time and within user budget for each case.

Reading suggestion: Should start with project Purpose and scope then by the paper arrangement.

Let's assume that Arab Academy wants to make an event to help 12th grade students know more about the academy and its colleges so the application will make the arrangement and will generate tickets till the maximum number of attendees is, what we can conclude from that story is the process of organizing the event made easier for both the academy and the students

1.4 Project Scope

This project, which will be a web-based application, will handle issues related to the scheduling and management of events within the institution. It will be confined to the creation of meetings by the event creator, and attendees can make reservations for the meetings and send information as regards the meeting to the event creator or the attendee, with detailed meeting descriptions such as time, duration, and agenda that will be needed for the meeting via their email and mobile phone. This system will also send a reminder before the meeting to remind the creator and attendees of the time and location of the meeting.

1.5 References

<https://www.momice.com/en/home>

<https://www.accelevents.com/blog/all-you-need-to-know-about-event-gamification>

<https://www.socialtables.com/blog/event-technology/event-management-tools/>

<https://www.nunify.com/blogs/hybrid-event-app/>

<https://www.vfairs.com/>

<https://sportevents.me>

<https://www.conventures.com>

<https://www.afoxevent.com>

<https://canapii.com/>

1.6 Product Perspective

Replacement for exciting products which are in the event management and so it has the same characteristics of the existing systems in addition to the gamification and that's what make the system unique with respect to geographical location.

Elements

- **Badges / Achievements** Player



Badges and achievements are a form of feedback. Award them to people for accomplishments.
Use them wisely and in a meaningful way to make them more appreciated.

- **Sharing Knowledge** Philanthropist



For some, helping other people by sharing knowledge with them is its own reward.
Build the in the ability for people to answer questions and teach others.

- **Voting / Voice** Disruptor



Give people a voice and let them know that it is being heard. Change is much easier if everyone is on the same page.

- **Innovation Platform** Disruptor



Disruptors think outside the box and boundaries of your system. Give them a way to channel that and you can generate great innovations.

- **Care-taking** Philanthropist



Looking after other people can be very fulfilling. Create roles for administrators, moderators, curators etc. Allow users to take a parental role.

- **Learning / New Skills** Achiever



What better way to achieve mastery than to learn something new? Give your users the opportunity to learn and expand.

- **Creativity Tools** Free Spirit



Allow people to create their own content and express themselves. This may be for personal gain, for pleasure or to help other people (teaching materials, levels, gear, FAQ etc).

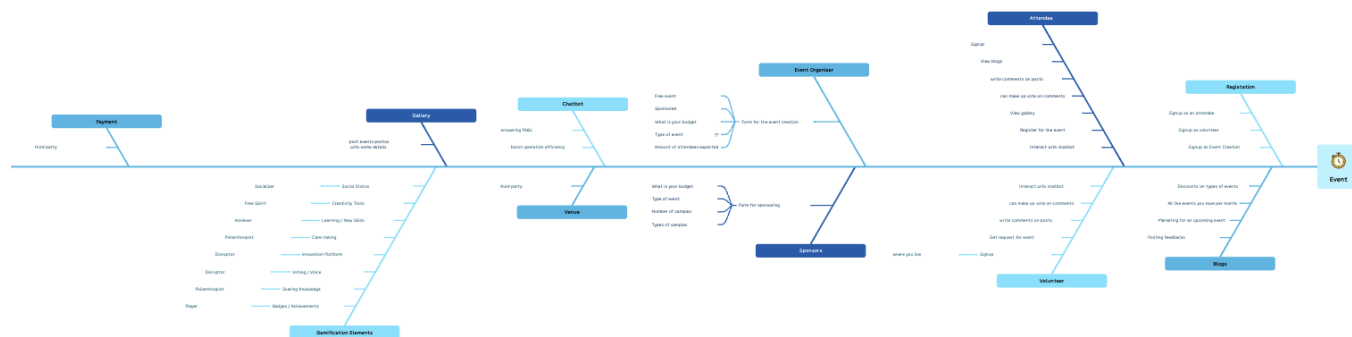
- **Social Status** Socializer



Status can lead to greater visibility for people, creating opportunities to create new relationships. It can also feel good. You can make use of feedback mechanics such as leaderboards and certificates.

1.7 Product Features

- Event creation and management: Event organizers will be able to create and manage events using the system. This includes setting up event details, such as the event name, date, location, and description.
- Participant registration and management: Participants will be able to register for events and manage their participation. This includes creating a profile, selecting events to participate in, and tracking their progress.
- Task creation and management: Event organizers will be able to create and manage tasks for participants to complete. This includes setting up task details, such as the task name, description, and reward.
- Task completion tracking: The system will track the completion of tasks by participants. This includes updating the participant's progress and providing feedback on task completion.
- Reward management: The system will manage the distribution of rewards to participants who complete tasks. This includes tracking the number of rewards earned by participants and providing the option to redeem rewards.



1.8 User Classes and Characteristics

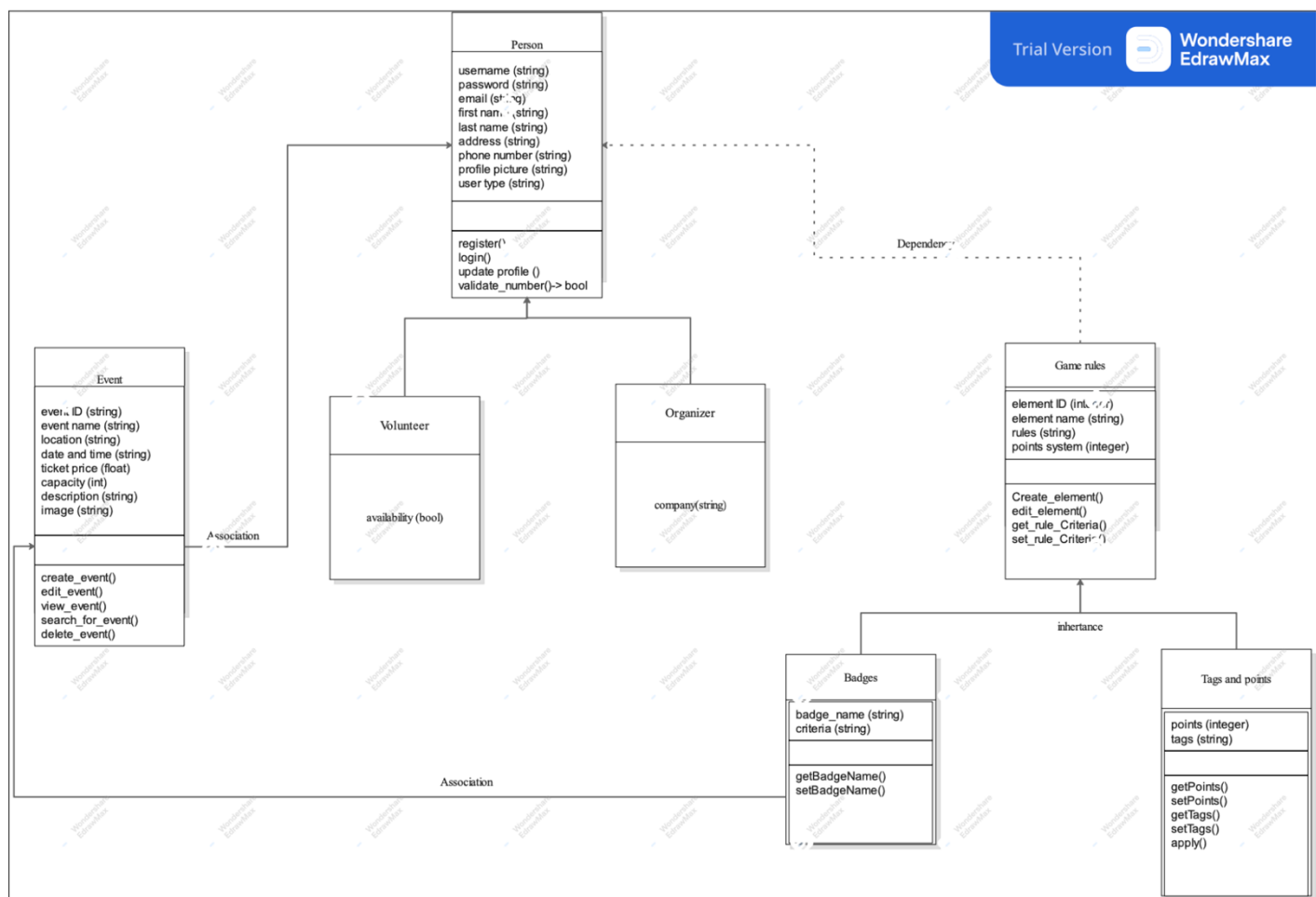
The gamified event management system will have four primary user classes:

Event Organizers :Who will create and manage events using the system. These users will have a higher level of access to the system and will be able to create and manage events and tasks.

Participants: Who will register for events, complete tasks, and earn rewards. These users will have a lower level of access to the system and will be able to view and track their progress.

Investor: Who will invest for a specific event wither by donating money or by providing needed material for an event in exchange for marketing for his brand as an example

Volunteer: Who will be in responsibility for organizing the event to make it seem perfect in exchange for a certificate as for example.



1.9 Operating Environment

The gamified event management system will be a web-based application and will run on a server. The system will require an internet connection to function and will be accessible from any device with a web browser. The following are the minimum operating environment requirements:

Operating System: The system will be accessible from any device that can connect to the internet and has a modern web browser installed, such as Google Chrome, Mozilla Firefox, Safari, or Microsoft Edge.

Hardware: The hardware requirements for using the system will be minimal, as the system will run on the server. Users will only need a device with an internet connection and a web browser.

Software: The system will be developed using modern web development technologies, including HTML, CSS, JavaScript, and React. The system will also utilize a database management system, MongoDB, to store and manage data.

Network: The system will require an internet connection to function. The speed and stability of the internet connection will affect the performance of the system. A stable and fast internet connection is recommended for optimal performance.

In addition, the system will be designed to be scalable and capable of handling a large number of users simultaneously. The system will utilize cloud-based hosting solutions to ensure that the system is always available and accessible to users. The system will also be designed with security in mind, utilizing encryption protocols to protect user data and prevent unauthorized access to the system.

1.10 Design and Implementation Constraints

1. Technology Constraints:

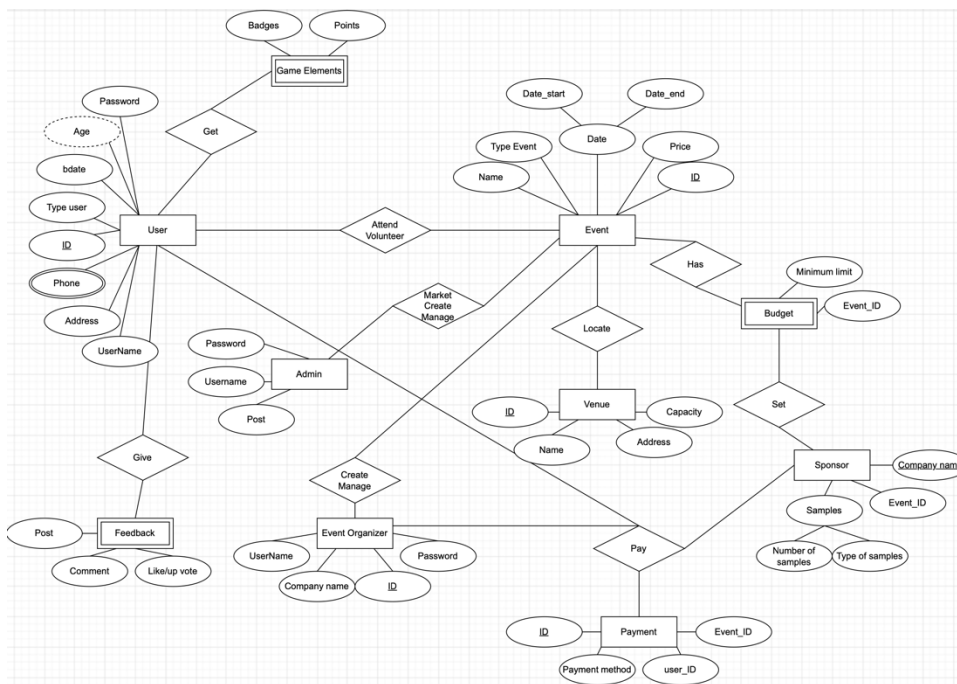
The system will be web-based and developed using MERN stack for the backend and HTML, CSS, and JS for the frontend.

2. Compatibility: The system is compatible with modern web browsers such as Google Chrome, Mozilla Firefox, Safari, and Microsoft Edge.

3. Security: The system must ensure data security by implementing secure authentication (auth0) and authorization mechanisms. User passwords is encrypted before storing them in the database.

4. Scalability: The system is designed to handle many users and events without compromising performance or stability.

5. Usability: The system user interface is easy to use even for people who have no experience dealing with event management systems.



1.11 User Documentation

You can choose either to register as a user or as an event creator and you will be given certain privileges based on your choice.

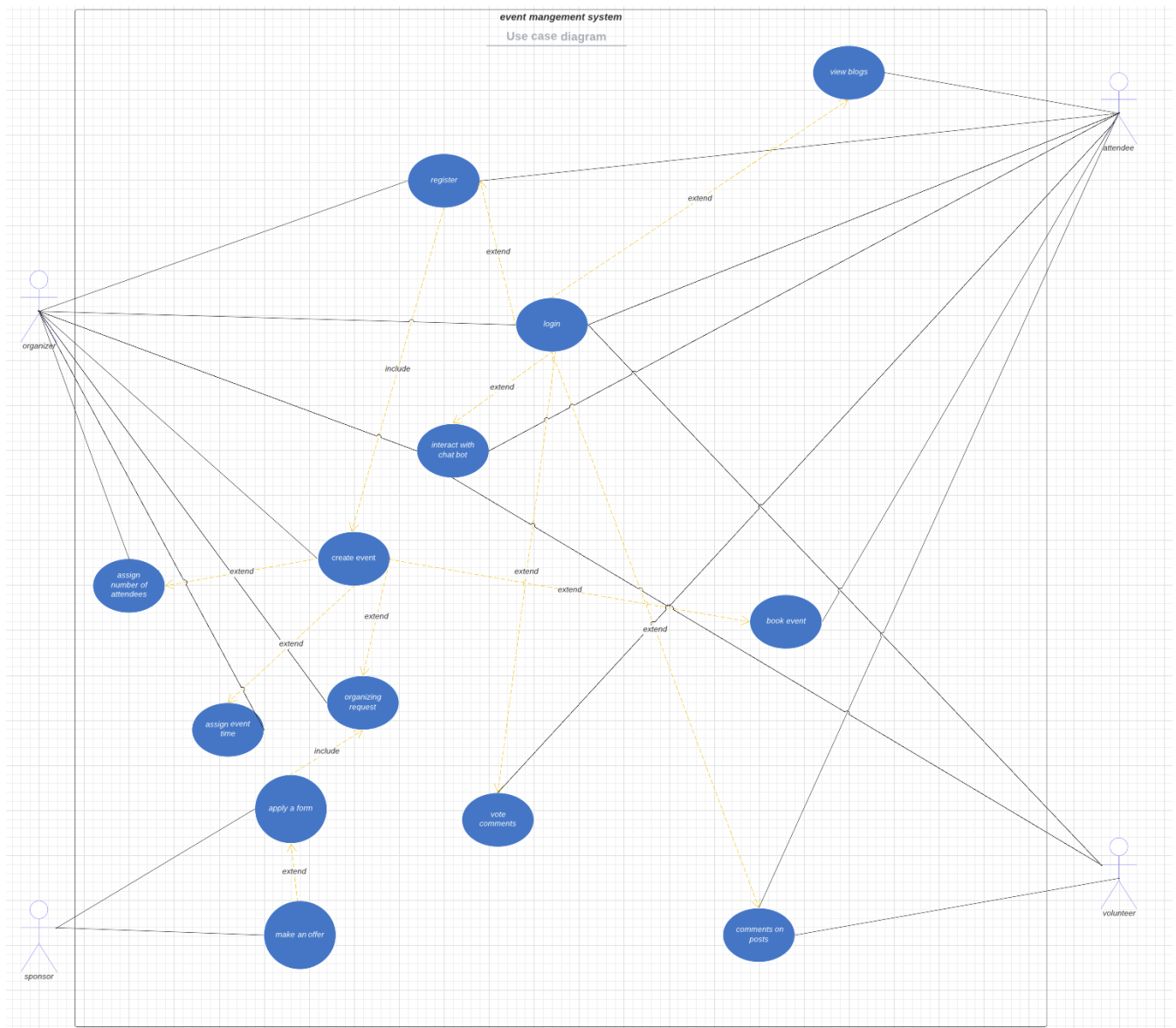
Where user can attend reserve or cancel while the event creator is responsible for creating the event and customizing it.

1.12 Assumptions and Dependencies

- 1) The Event Management System is designed for medium-sized organizations.
- 2) The System can handle up to 5000 attendees.
- 3) The System is accessible via modern browsers such as Google Chrome, Mozilla Firefox.
- 4) The System requires a stable internet connection.
- 5) The System will be hosted on a cloud server.
- 6) The System can integrate with third-party payment gateways.
- 7) The System can send email notifications automatically.
- 8) The System uses HTTPS protocol for secure communication between client-server.

2. System Features

The event management system creates, modifies, and chooses the maximum number of attendees of course that is a **high priority** because doing all those tasks on your own takes a lot of time and is difficult.



functional requirements :

- 1.Ability to register in the system.
- 2.Ability to save registration data in the database.
- 3.Ability to provide a secure login system with appropriate authentication measures in place for user accounts ^[1]_{SEP} management and data protection purposes.
- 4.Ability to create events , Manage them with required information such as date , time , location and type of the event
- 5.Ability to assign minimum and maximum number of attendees.
- 6.Ability to interact with chatbot
7. Ability to accept payments from event organizers to create the event.
8. Ability to accept payments from users for registering for events and purchasing the tickets online. ^[1]_{SEP}
9. Ability to send notifications and reminders about upcoming events.
10. Ability to generate reports about events for tracking performance and analytics.
11. Ability to send gamified badges for users

2.1 System Feature 1

3.1.1 Description and Priority

Save the event time in the database (high priority)

Sending the points *use of gamification* (medium priority)

Send notifications for users when similar event to what they are interested (medium priority)

3.1.2 Stimulus/Response Sequences

Search for an event using a topic name

System displays all events with that topic

User choose the event he wants to attend

Pay for it

Receive ticket

Attend the event

3.1.3 Functional Requirements

Webpage.

To make it easy for users to access it from any device with any OS.

Database:

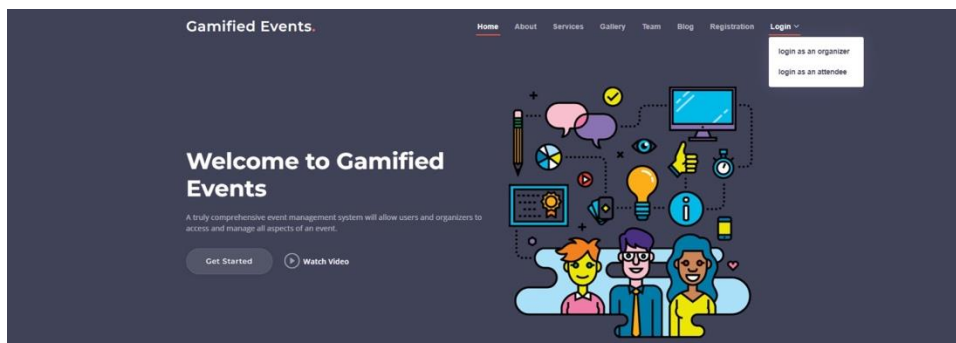
To save the users data and the events data and retrieve data when needed.

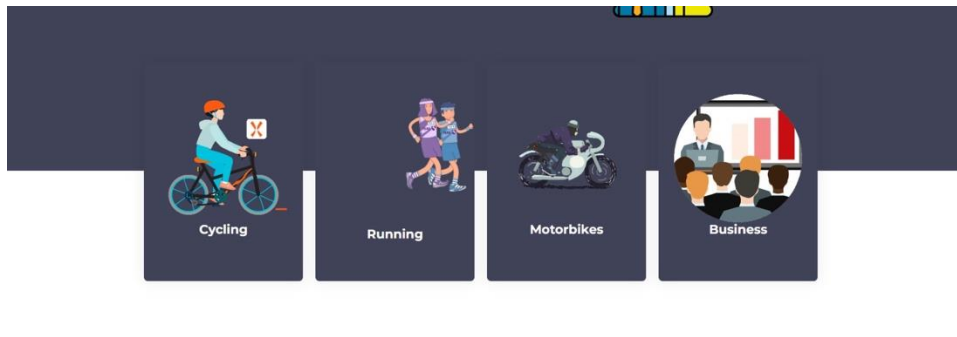
Gamification elements:

Will be sent with the user data in the DB to be retrieved and further used by the user.

3. External Interface Requirements

3.1 User Interfaces





Gamified Events.

Home **About** Services Gallery Team Blog Registration Login

About Us

An event management system essentially helps you keep track of every detail related to an event—both large and small—and can help you collaborate with vendors, partners, and team members to execute a successful event.

Create an event and manage all its aspects



Depending on your needs, budget, and event size, you might find that one program works better than others. To cover your bases, here is our software for event planning that covers most of the main event plan:

- Venue.
- Budget.
- Sponsors.
- Marketing.
- Volunteers.
- Equipment.
- Attendees.



Gamified Events

Home

About

Services

Gallery

Team

Blog

Registration

Login

Our Services



Lease a Venue

Our website partners with top-notch third-party venues to offer a seamless and stress-free experience for event organizers. The process of leasing a venue is made simple, with a wide range of options to choose from.

[Read more ->](#)



Marketing

Our website provides a platform for event organizers to market their events via blogs. This helps to reach a wider audience and generate more buzz about the event. Attendees can also share their feedback and experiences through the blog, providing valuable insights for event organizers.

[Read more ->](#)



Easy payment

Our website offers a hassle-free payment experience, with options to pay through third-party payment providers and e-wallets. This makes the payment process smooth and convenient for event organizers and attendees alike.

[Read more ->](#)



Gallery

Our website features a photo gallery where attendees can view photos of the event and relive their experience. This helps to create a lasting memory for attendees and provides valuable marketing material for event organizers.

[Read more ->](#)



Mange expenses

Our website offers a budget setting feature that allows event organizers to set a budget before the event. This helps to keep the event organized and on track, ensuring that all expenses are managed effectively.

[Read more ->](#)



ChatBot

Our website features a cutting-edge chatbot that helps event organizers and attendees with all their queries and concerns. The chatbot is available 24/7 and offers quick and efficient assistance.

[Read more ->](#)

Gamified Events

Home

About

Services

Gallery

Team

Blog

Registration

Login

Pricing



Free Plan

\$0 / month

- ✓ Basic photo gallery
- ✓ Basic marketing support (blog only)
- ✓ Access to basic event management features
- ✓ Basic chatbot support

✗ Venue options

[Buy Now](#)



Organizer Plan

\$29 / month

- ✓ Access to advanced event management features
- ✓ Advanced marketing support (blog + social media)
- ✓ Budget setting feature
- ✓ Advanced chatbot support

[Buy Now](#)



Sponsor Plan

\$49 / month

- ✓ Access to premium event management features
- ✓ Exclusive venue options
- ✓ Premium marketing support (blog + social media + targeted ads)
- ✓ Priority budget setting feature

[Buy Now](#)

3.2 Hardware Interfaces

Android Phone, iPhone, Mac, PC

In short, this website can work on any smart phone connected to the internet or any computer ...
IF The device browser supports html & JavaScript

Supported versions:

3.3 Software Interfaces

The Gamified event management system requires the following software interfaces:

1. Operating System: The system is compatible with all major operating systems such as Windows, Mac OS X, Linux, Android and IOS.
2. Web Server: The backend of the system is built using MERN stack (MongoDB, Express.js, React.js, and Node.js) which requires Node.js installed on the server.
3. Database Management System: The database used for storing user data is MongoDB.
4. Front-end Interface: The front-end of the system is built using HTML5, CSS3, and JavaScript which are compatible with all modern web browsers.
5. Third-Party APIs: The system integrates with third-party APIs for features such as payment processing, email notifications and social media integration.

3.4 Communications Interfaces

The site will use the form of HTTPS for optimized security.

Data exchange: Client sends a request to the server to retrieve a list of all events
Server responds with a list of all events in JSON format

Client sends a request to the server to create a new event

Server responds with a success message and the created event in JSON format:

4. Other Nonfunctional Requirements

4.1 Performance Requirements:

- 1.Ensure that the system is scalable enough so that it can handle required volumes of data or multiple concurrent users without any performance issues or delays in response times.
- 2.Ensure that the system is reliable enough so that it can be used consistently without any unexpected errors occurring when hosting events on it.
- 3.Ensure that the system has an intuitive interface so that users can easily navigate through it easily.
- 4.Ensure that the system is compatible with various operating systems, browsers, devices, etc., so that users can access it without any compatibility issues regardless of their setup.

4.2 Safety Requirements

- 1.Data Encryption: Ensure sensitive data such as user information, payment details, and event details are encrypted before they are stored in the database.

2. Input Validation: Validate all inputs from users to prevent malicious attacks such as SQL injection, XSS, and CSRF.

4.3 Security Requirements

The users must authenticate themselves to use the system.

Only the event creator can modify his event.

The user is the only person who can cancel the request but also a user can buy more than tickets for others

If an event creator has been delayed for more than one time (5 minutes or more) he will not be authenticated to make any further events.

Added Auth0 authentication and verification

4.4 Software Quality Attributes

Compatibility:

The software should be compatible with various devices such as:

1. Desktop computers running Windows, MacOS or Linux operating systems.
2. Mobile devices such as smartphones and tablets running iOS or Android operating systems.
3. Web browsers such as Google Chrome, Mozilla Firefox, Safari, and Microsoft Edge.

Gamification:

Quality attributes related to gamification should include the following:

1. Engagement: The system should be able to engage users in various activities through game-like features such as points, badges, leaderboards, and challenges.
2. Fun factor: The system should provide a fun and enjoyable experience for users by incorporating elements of playfulness and excitement into the design.
3. User feedback: The system should provide feedback to users on their progress and performance in various activities to motivate them to continue participating.

4. Personalization: The system should allow users to customize their profiles and preferences to make the experience more personal and relevant.

5. Social interaction: The system should enable social interaction among users through features such as chat rooms, forums, or social media integration.

6. Rewards: The system should offer rewards such as discounts, coupons or special privileges for active participation in events or activities organized through the system.

7. Accessibility: The system should be accessible to all users regardless of their physical abilities or limitations.

8. Security: The system should ensure that user data is kept secure and protected from unauthorized access or misuse by implementing appropriate security measures such as encryption and authentication protocols.

5. Other Requirements

organization Department approval.

The approval of the licensing center.

The approval of the competent civil protection department. Tax card.

Commercial Register.

Title deed, lease or possession of the project headquarters. Proof of identity for the project owner.

Appendix A: Glossary

The event management system: used to manage and organize the various aspects of event planning, including scheduling, budgeting, and resource management.

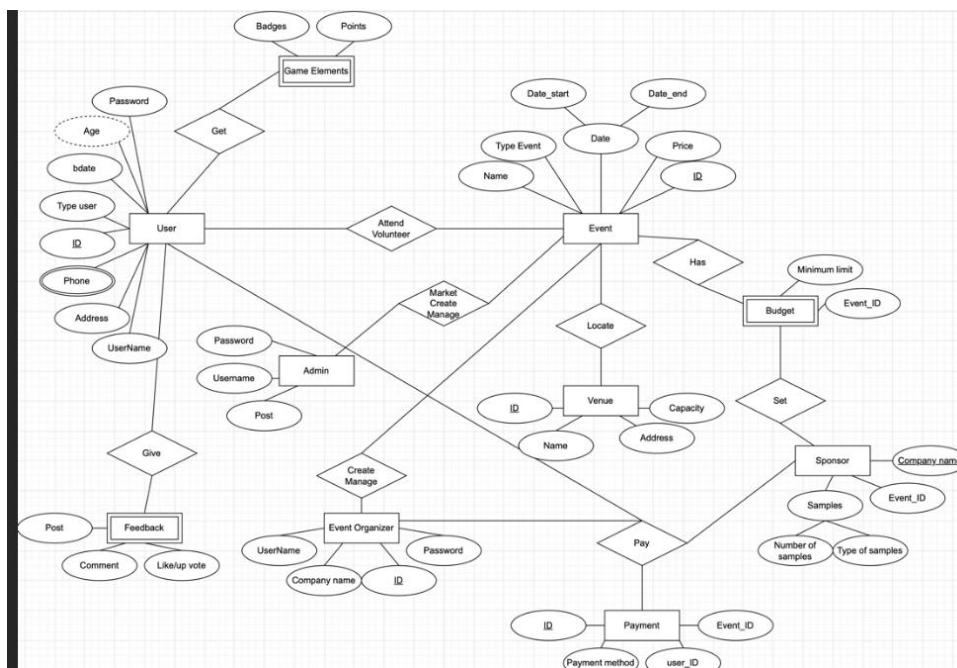
Event organizers: A person who is responsible for creating the event and assign maximum, minimum number of attendees

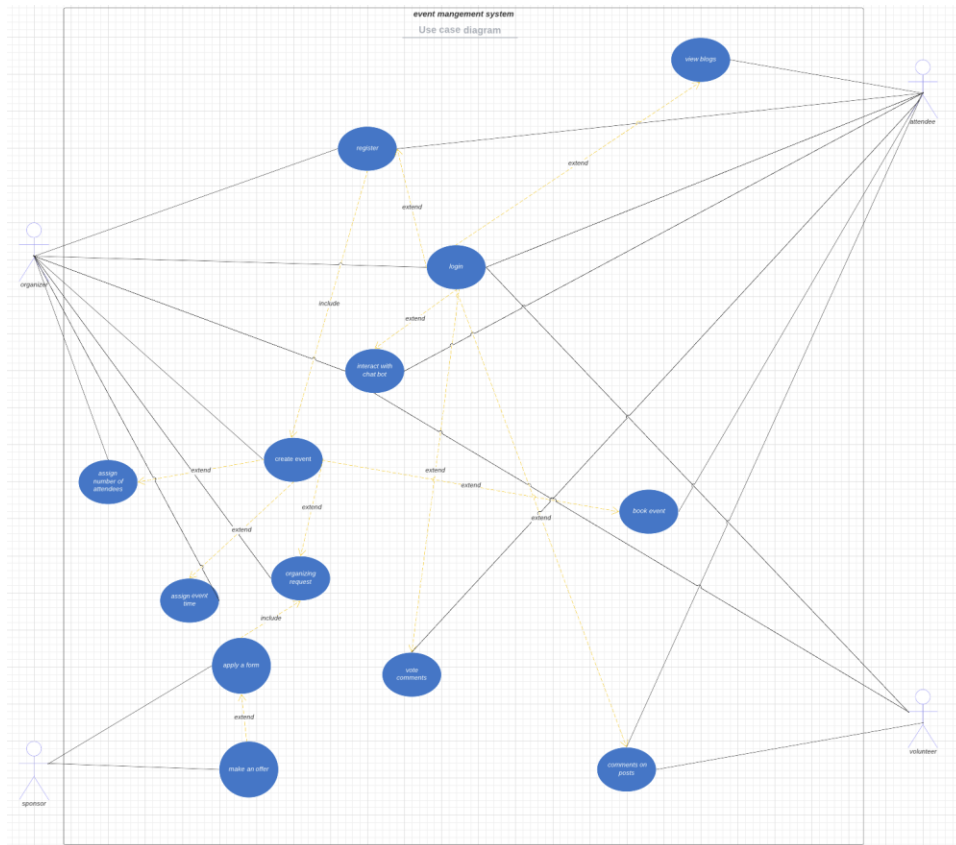
volunteer: A person responsible for organizing and coordinating events.

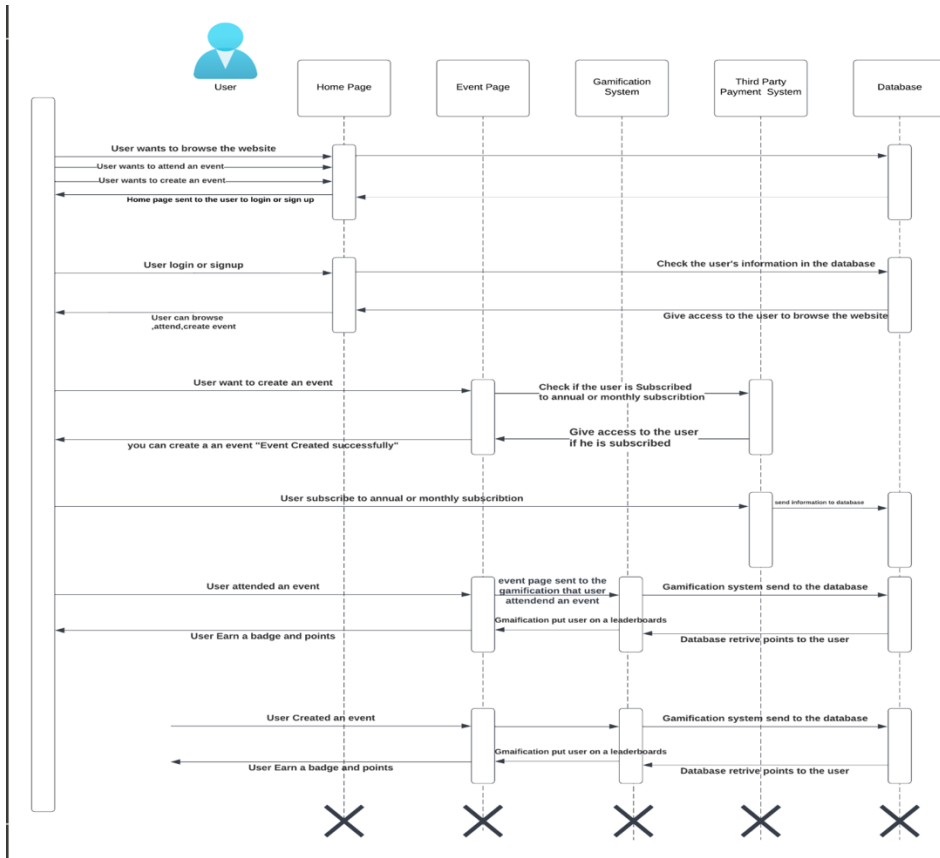
Sponsor: A person who may be interested in sponsoring a specific event.

Budgeting: The process of allocating funds to several aspects of an event to ensure it runs smoothly and within budget.

Appendix B: Analysis Models







Use Case Name	Log in
Summary	This use case describes how a user logs into the gamified event management system to prove their identity and access their account.
Actors	User
Preconditions	None
Description	1. The user selects the "log in" function from the main menu. The system prompts the user to enter their username and password. The user enters their username and password. The system verifies the user's credentials and logs them into the system. If the user has earned new badges or rewards, the system displays a notification.
Alternatives	Sign up, If the user's credentials are not valid, the system prompts the user to try again or reset their password.
Postcondition	The user is logged into the system and can access their account.
Dependencies	Login

Use Case Name	Create Event
Summary	This use case describes how an event organizer creates a new event in the system.
Actors	Event Organizer
Preconditions	The Event Organizer is logged in to the system.
Description	1. The Event Organizer selects the "Create Event" option from the main menu.
	2. The system displays the "Create Event" form.
	3. The Event Organizer fills in the form with event details, including event name, date, time, location, description, and any other relevant information.
	4. The Event Organizer submits the form.
	5. The system saves the new event in the database and assigns a unique event ID to it.
Postconditions	A new event is created and added to the system.
Dependencies	Log in

Use Case Name	Register for Event
Summary	This use case describes how a system user registers for an event in the gamified event management system.
Actors	System User
Preconditions	The user registered in the site
Description	1. The system user selects the desired event from the events list. The system displays the event details page. The system user clicks on the "Register" button. The system registers the user for the event and increments the user's event attendance counter. The system updates the user's badge status based on the number of events attended.
Alternatives	Previously had an account
Postconditions	The user is registered for the event, badge status is updated.
Dependencies	Login

Use Case Name	View Event Attendance
Summary	This use case describes how an event organizer views the attendance details of their events in the gamified event management system.
Actors	Event Organizer
Preconditions	The event organizer has created at least one event.
Description	1. The event organizer selects "My Events" from the main menu. The system displays a list of events created by the organizer. The event organizer selects the desired event. The system displays the attendance details for the selected event, including the number of attendees and their names.
Alternatives	Sign up
Postconditions	Previously had an account, The event organizer views the attendance details for their event.
Dependencies	Logged in as event organizer

User	Organizer
1. Open event management website	1. Open event management website
2. Browse or search for events	2. Create an account using email address
3. Select event for more information	3. Receive confirmation code (password) on email
4. Create account using email address	4. Confirm login validity by entering confirmation code
5. Receive confirmation code (password) on email	5. Create a new event
6. Confirm login validity by entering confirmation code	6. Fill in event details such as name, date, location, description, and pricing
7. Select number of tickets to purchase	7. Set up payment options and ticket sales
8. Add tickets to cart	8. Publish the event and make it available for users to browse and purchase tickets
9. Proceed to checkout and confirm purchase	-
10. Receive purchased ticket(s) by email	-